Machine Learning Tutorial

Title: How Neural Network Architecture Affects Performance: A Study of Depth vs Width in Multilayer Perceptron's

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Introduction

Neural networks are one of the most powerful and widely used techniques in modern machine learning. Inspired by the structure of the human brain, they are capable of learning complex patterns from data and are widely applied in tasks such as image recognition, natural language processing, and predictive analytics.

Among the different types of neural networks, the Multilayer Perceptron (MLP) is one of the most fundamental types of neural networks. Even though the MLP is considered simple, it is the foundation of many complex deep learning models. One of the major challenges in using neural networks is the selection of the appropriate architecture.

Two critical aspects of neural network architecture are **depth** and **width**:

- **Depth** refers to the number of hidden layers in the network.
- **Width** refers to the number of neurons in each hidden layer.

These architectural choices significantly impact how well a neural network learns from data. A network that is too simple may fail to capture patterns (underfitting), while a network that is too complex may memorise the data instead of generalising (overfitting).

In the tutorial, the effect of the depth and width of an MLP on its performance will be examined. The experiments will be conducted on a non-linear dataset, and the results will be analysed through visualizations. The objective of the research is to achieve an intuitive understanding of the effect of the neural network's architecture on the learning outcomes.

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2. Key Concepts

2.1 Multilayer Perceptron (MLP)

The Multilayer Perceptron is a feed-forward neural network with the following three parts:

- The input layer, which receives the data
- The hidden layers, which process the data

- The output layer, which makes the prediction

Each neuron in the MLP calculates the weighted sum of the inputs and then applies the activation function to produce the output. The backpropagation algorithm is used to train the MLP and adjust the weights to achieve the minimum prediction error.

2.2 Depth of a Neural Network

The depth of the network represents the number of hidden layers used. Networks with one hidden layer are termed shallow networks, whereas networks with more than one hidden layer are termed deep networks.

The depth of the network can be increased to allow the network to learn more complex features. For example, when dealing with classification networks, deeper layers can be used to represent more abstract features. However, deeper networks can be more computationally expensive to train.

2.3 Width of a Neural Network

The width of the network is the number of neurons in each of the hidden layers. The wider the network, the more neurons each of the hidden layers will have, which enables the network to recognize more complex patterns in the data.

Increasing the width of the network improves the capacity of the model, but the number of parameters increases, which might lead to overfitting, especially if the data is insufficient.

2.4 Underfitting and Overfitting

- **Underfitting** occurs when a model is too simple to capture the underlying structure of the data.
- **Overfitting** occurs when a model learns the training data too well, including noise, leading to poor generalisation.

Achieving the right balance between depth and width is essential to avoid both issues.

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3. Methodology

3.1 Dataset

To assess the effects of neural network architecture, a synthetic data set is created with the use of the "make_moons" function from Scikit-learn. This data set contains two interweaving half-moon shapes and is thus nonlinearly separable.

This data set offers a clear visual representation of learning complex decision boundaries.

3.2 Data Preparation

The data set was divided into sets:

- 80% for training
- 20% for testing

This data set division helps to ensure that the data set is tested on unseen data, thereby giving a true measure of the performance of the data set.

3.3 Model Configurations

Three different neural network architectures were tested:

Model 1: Shallow Network

- 1 hidden layer
- 10 neurons

Model 2: Medium Network

- 2 hidden layers
- 20 neurons per layer

Model 3: Deep Network

- 4 hidden layers
- 50 neurons per layer

All models were implemented using the MLPClassifier from Scikit-learn with the same training parameters to ensure a fair comparison.

3.4 Evaluation Metrics

The models were evaluated using:

- **Accuracy**: Measures the proportion of correctly classified samples.
- **Visualisation**: Decision boundary plots were used to understand how each model separates the data.

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4. Results

4.1 Accuracy Comparison

The results indicated a trend in the performance of the models:

- The performance of the shallow network was low in terms of accuracy, indicating low learning capabilities.

- The performance of the medium network was better, as it struck a balance between complexity and generalization.
- The performance of the deep network was the best but indicated some signs of overfitting.

This shows that the more complex the model, the better the performance, but only to a certain extent.

4.2 Decision Boundary Analysis

Decision boundary plots provided deeper insights into model behaviour:

- **Shallow Network**:
    - Produced a simple, nearly linear boundary
    - Failed to capture the curved structure of the data
    - Demonstrated underfitting
- **Medium Network**:
    - Produced smooth and accurate boundaries
    - Successfully captured the non-linear pattern
    - Showed good generalisation
- **Deep Network**:
    - Produced highly complex boundaries
    - Closely followed the training data
    - Risk of overfitting observed

4.3 Training Behaviour

The training loss curves showed that the models learned in the following manner:

- Shallow models had fast convergence rates but higher loss values
- Medium models had stable convergence
- Deep models took more iterations and sometimes fluctuated in learning

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5. Discussion

The results clearly show that architecture plays a vital role in the performance of a neural network.

5.1 Impact of Depth

Increasing depth enables the network to:

- Learn hierarchical representations
- Capture more complex relationships

However, increasing depth also makes the network:

- Need more data
- More difficult to train
- More likely to overfit unless regularised

5.2 Impact of Width

Increasing the number of neurons:

- Enhances the model's ability to represent complex functions
- Improves performance on non-linear datasets

However:

- Too many neurons increase model complexity
- Overfitting becomes more likely

5.3 Trade-Off Between Simplicity and Complexity

The results demonstrate a key principle:

- Increasing complexity improves learning up to a point, beyond which it harms generalisation.

The medium network provided the best balance between accuracy and stability, making it the most effective model in this experiment.

5.4 Practical Implications

When designing neural networks:

- Start with a simple model
- Gradually increase complexity
- Use validation data to monitor performance
- Apply regularisation techniques if needed

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6. Ethical Considerations

Neural networks are widely used in real-world applications such as healthcare, finance, and recruitment. While they are powerful, they also raise important ethical concerns.

- **Bias**: Models may learn biases present in training data
- **Transparency**: Neural networks are often considered "black boxes"
- **Over-reliance**: Automated decisions may impact human lives

It is important to ensure that models are trained responsibly and evaluated for fairness and reliability.

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7. Conclusion

This tutorial explored how the depth and width of a neural network affect its performance using a Multilayer Perceptron.

Key findings include:

- Shallow networks are limited in learning complex patterns
- Deep networks can model complex relationships but risk overfitting
- Medium-sized networks often provide the best balance

The study highlights the importance of carefully selecting neural network architecture based on the problem and dataset.

Understanding these trade-offs enables practitioners to design more effective and reliable machine learning models.

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8. References

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